(a) Shared: identical for both agents



Camera frame: what the agent sees
224 × 224 colour image

DINOv2
frozen

17 tokens × 384
= 6,528 numbers

Observation vector
6,539 numbers

image features

proprio

Proprioception: 11 numbers

on ground	0 (airborne)
yaw / pitch	0.69 (125°) / 0.30 (27°)
velocity (y, x, z)	0, 0, 0
position vs. spawn	−1.2, 1.0, 8.8
food / health	1.0 / 1.0

11 numbers
(skips DINOv2)

real example, same time step from a recorded episode

bands not drawn to scale

(b) Internal model: the one difference

Input layers

Feedforward layer

PPO
no memory
learns:
on-policy rollouts,
GAE, clipped loss

Input layers

RSSM world model

Dreamer
recurrent memory
learns:
fit model on replay,
policy in imagination

(c) Outputs: same

Actor head

action
1 of 12: move, turn,
jump, attack, . . .

same architecture in both agents

Critic head

value V
expected future reward